SL Math School: Random Growth Models, etc.
Tutorial 8: Stopping Domains

1 Prologue

Consider any (random) line ensemble $\{\mathcal{L}_n(x)\}_{n\in J,\, x\in I}$ satisfying the Brownian Gibbs Property. That is, for a rectangle $R = \{k_1, \dots, k_2\} \times [a,b] \subset J \times I$, we define the external σ-algebra $\mathcal{F}_{\text{ext}}^{(R)}$ as

$$\mathcal{F}_{\text{ext}}^{(R)} = \sigma\left(\mathcal{L}_i(x) \,:\, i \notin \{k_1, \dots, k_2\} \text{ or } x \notin (a,b)\right).$$

The Brownian Gibbs Property then states the law of $\mathcal{L}$ restricted to R, conditioned on everything outside of R, is equal in law to non-intersecting Brownian Bridges with appropriate boundary conditions. So far, we have only considered the case where the time interval has been deterministic. Today we will consider the case where the interval is random as well.

Definition: Consider two random variables ℓ, τ taking values in I. We say that the random interval $[\ell, \tau]$ is a **stopping domain** if almost surely $\ell \leq \tau$, and for all $a < b \in I$

$$\{\ell \leq a,\, \tau \geq b\} \in \sigma\left(\mathcal{L}_i(x) \,:\, i \in J,\, x \notin (a,b)\right).$$

That is, the values of ℓ, τ are measurable with respect to the sigma-fields generated by the ensemble outside the random interval (ℓ, τ). Intuitively, I like to think of stopping domains as analogous to stopping times; ℓ only depends on the information to the left of it, and τ only depends on the information to the right.

Lemma (Strong Domain Markov Property): Suppose that $[\ell, \tau]$ is a stopping domain. Consider any interval $\{k_1, k_1+1, \dots, k_2\} \subset J$ and consider the random rectangle

$$R = \{k_1, \dots, k_2\} \times [\ell, \tau].$$

Then we have the following strong Brownian Gibbs Property for R:

$$\begin{aligned}\mathbb{P}\left(\left(\mathcal{L}_{k_1}, \dots, \mathcal{L}_{k_2}\right)|_{[\ell,\tau]} \in \cdot \,\middle|\, \mathcal{F}_{\text{ext}}^{(R)}\right) &= \mathcal{B}_{k;\overline{x},\overline{y}}^{[\ell,\tau]}\left(\cdot \,\middle|\, \text{NoTouch}_{k,[\ell,\tau]}^{f,g}\right) \\ &:= \mathbb{P}_{\text{NI}}^{[\ell,\tau],\overline{x},\overline{y},f,g}(B \in \cdot)\end{aligned}$$

where now

$$\begin{aligned}\overline{x} &= \left(\mathcal{L}_{k_1}(\ell), \mathcal{L}_{k_1+1}(\ell), \dots, \mathcal{L}_{k_2}(\ell)\right), \\ \overline{y} &= \left(\mathcal{L}_{k_1}(\tau), \mathcal{L}_{k_1+1}(\tau), \dots, \mathcal{L}_{k_2}(\tau)\right).\end{aligned}$$

and

$$f = \begin{cases} \mathcal{L}_{k_1-1}, & k_1 - 1 \in J, \\ +\infty, & k_1 - 1 \notin J, \end{cases}$$

$$g = \begin{cases} \mathcal{L}_{k_2+1}, & k_2 + 1 \in J, \\ -\infty, & k_2 + 1 \notin J. \end{cases}$$

2 Problems

2.1 Core Problems

Problem 1 (Which Random Intervals are Stopping Domains?)**.** Let $\mathcal{L} = (\mathcal{L}_1, \ldots, \mathcal{L}_n)$ be a continuous line ensemble on $[-10, 10]$. Which of the following are stopping domains?

(a)

$$\begin{aligned} \ell &= \inf\{x \in [-10, -1] : \mathcal{L}_1(x) - \mathcal{L}_2(x) \leq 1\}, \\ \tau &= \sup\{x \in [1, 10] : \mathcal{L}_1(x) - \mathcal{L}_2(x) \leq 1\}. \end{aligned}$$

(b)

$$\begin{aligned} \ell &= \sup\{x \in [-10, 0] : \mathcal{L}_1(x) - \mathcal{L}_2(x) \leq 1\}, \\ \tau &= \inf\{x \in [0, 10] : \mathcal{L}_1(x) - \mathcal{L}_2(x) \leq 1\}. \end{aligned}$$

(c)

$$\begin{aligned} \ell &= \operatorname{argmax}_{x \in [-10,10]} \big(\mathcal{L}_1(x) - \mathcal{L}_2(x)\big), \\ \tau &= 10. \end{aligned}$$

(d)

$$\begin{aligned} \ell &= \inf\left\{x \in [-10, 10] : \min_{1 \leq i \leq k-1} \big(\mathcal{L}_i(x) - \mathcal{L}_{i+1}(x)\big) \leq 1\right\}, \\ \tau &= \sup\left\{x \in [-10, 10] : \min_{1 \leq i \leq k-1} \big(\mathcal{L}_i(x) - \mathcal{L}_{i+1}(x)\big) \leq 1\right\}. \end{aligned}$$

Problem 2 (No Big Max)**.** Let $\mathcal{L} = (\mathcal{L}_1, \ldots, \mathcal{L}_n)$ be a line ensemble satisfying the Brownian Gibbs property on $[-2, 2]$, ordered so that

$$\mathcal{L}_1(x) > \mathcal{L}_2(x) > \cdots > \mathcal{L}_n(x)$$

for all $x \in [-2, 2]$. Let $t > 0$. Prove that

$$\mathbb{P}\left(\sup_{x \in [0,2]} \mathcal{L}_1(x) \geq t\right) \leq 2\mathbb{P}\left(\mathcal{L}_1(0) \geq \frac{t}{4}\right) + \mathbb{P}\left(\mathcal{L}_1(-2) < -\frac{t}{2}\right).$$

Hint: Consider the stopping domain $[-2, \chi]$ where

$$\chi = \sup\{x \in [0, 2] : \mathcal{L}_1(x) \geq t\}.$$

2.2 Extra Challenge Problem

Problem 3 (Dropping Boundary Conditions)**.** Fix a compact interval $I \subset \mathbb{R}$ and boundary data $(\mathbf{x}, \mathbf{y}, f, g)$. Suppose that

$$B = (B_1, \ldots, B_n) \sim \mathbb{P}_{\mathrm{NI}}^{I,\mathbf{x},\mathbf{y},+\infty,-\infty}.$$

Suppose that

$$\mathbb{P}\left(B_1(r) < f(r) \text{ and } B_n(r) > g(r) \text{ for all } r \in I\right) \geq \frac{1}{C}.$$

Show that

$$\frac{d\mathbb{P}_{\mathrm{NI}}^{I,\mathbf{x},\mathbf{y},f,g}}{d\mathbb{P}_{\mathrm{NI}}^{I,\mathbf{x},\mathbf{y},+\infty,-\infty}} \leq C.$$

Problem 4 (Lower Point Control)**.** Let $\mathcal{L} = (\mathcal{L}_n)_{n\in\mathbb{N}}$ be a line ensemble satisfying the Brownian Gibbs Property, with $\mathcal{L}_i : \mathbb{R} \to \mathbb{R}$. Define the parabolically shifted ensemble

$$\mathcal{A}_i(x) = \mathcal{L}_i(x) + x^2.$$

Assume that the shifted ensemble $\mathcal{A}$ is stationary in x. Also assume that there exist constants $c, d > 0$ so that for all $r \geq 0$,

$$\mathbb{P}\left(|\mathcal{A}_1(0)| > r\right) \leq ce^{-dr^{3/2}}.$$

(a) Show that there exist constants $c, d > 0$ so that, for all $T \geq 1$,

$$\mathbb{P}\left(\sup_{x\in[-T/2,T/2]} \mathcal{A}_2(x) \leq -\frac{5}{8}T^2\right) \leq ce^{-dT^3}.$$

(b) Show that there exist constants $a, b > 0$ so that, for all r large enough,

$$\mathbb{P}\left(\mathcal{L}_2(0) \leq -r\right) \leq ae^{-br^{3/2}}.$$

(c) Conclude that for every $k \in \mathbb{N}$, there exist constants $c_k, d_k > 0$ so that, for all $r \geq 0$,

$$\mathbb{P}\left(|\mathcal{A}_k(0)| > r\right) \leq c_k e^{-d_k r^{3/2}}.$$

Hint: For part (a), argue that if $\mathcal{A}_2$ stays below $-\frac{5}{8}T^2$ on $[-T/2, T/2]$, then after resampling the first curve on this interval, the first curve has a positive probability of sagging below its typical parabolic height. Use the one-point tail bound for $\mathcal{A}_1(0)$.

For part (b), use stationarity and part (a) as well as the stopping domain $[\ell, \tau]$ given by

$$\ell = \inf\left\{x \in [-2T, -T] : \mathcal{A}_2(x) > -\frac{5}{8}T^2\right\},$$
$$\tau = \sup\left\{x \in [T, 2T] : \mathcal{A}_2(x) > -\frac{5}{8}T^2\right\}.$$

Choose $T = \epsilon r^{1/2}$ for a sufficiently small constant $\epsilon > 0$, so that the endpoint values $\mathcal{L}_2(\ell)$ and $\mathcal{L}_2(\tau)$ are both at least $-r/2$. Finally, use the Brownian bridge cost for a curve with endpoints at least $-r/2$ on an interval of length $O(T)$ to dip below $-r$ at 0.